GOUTHAM SHIMOGA UMESH

(602) 348-6090 | gouthamshimogaumesh@gmail.com | LinkedIn | Portfolio | F-1 OPT

EXPERIENCE

Assistant Research Technologist | Arizona State University, Tempe, AZ *Dec 2025 – Present*

- Conducted VOF multiphase CFD simulations on 8 nozzle geometries in ANSYS Fluent, developed custom field functions, and post-processed contour data in MATLAB to extract spray metrics and identify 2 optimal configurations for target operating conditions.
- Built a standardized simulation template cutting per-geometry setup time by ~70%, enabling all 8 geometries to be processed in under 2 weeks against a projected 5+ week timeline.
- Evaluated FDM-printed nozzle prototypes against CFD predictions post-build, recording dimensional and performance deviations to close the loop between analytical models and physical hardware.

Design Engineering Intern | Hindustan Aeronautics Limited (HAL), Aircraft R&D Center, Bengaluru, India *Mar – May 2023*

India's State-Owned Aerospace and Defense Manufacturer (Equivalent to Boeing and Lockheed Martin)

- Modeled a parametric 3D semi-span wing assembly in CATIA V5 incorporating 15 ribs, dual-spar construction, taper ratio 0.45, and aluminum alloy skin panels across a 29.1 m span, managing complex surface geometry to meet a hard 530 kg per-wing mass budget.
- Performed conceptual sizing from first principles, deriving 12 performance constraint curves across 5 parameters and 6 FAR Part 25 climb segments to determine optimal wing loading and power loading at the design point.
- Validated wing structure under 2.5g maneuvering loads in ANSYS Mechanical, achieving a margin of safety of 0.84 with tip deformation at 3.1% of half-span, confirming all structural limits through hand calculations and FEA.
- Collaborated directly with aerodynamics, stress, and manufacturing teams to reconcile competing design constraints, communicating analysis findings across 3 engineering groups to drive design decisions.

Manufacturing Engineering Intern | Hindustan Aeronautics Limited (HAL), Helicopter Division, Bengaluru, India *Mar – Apr 2023*

Military Aircraft Production Facility, Advanced Light Helicopter and Light Combat Helicopter Programs

- Supported final assembly of 3 simultaneous military helicopters (2x ALH for the Indian Coast Guard, 1x LCH for the Indian Army), rotating across 7 stations covering fuselage integration, rotor installation, and systems routing.
- Flagged a 4 mm landing-gear symmetry deviation on ALH-2 during routine dimensional verification (±2 mm tolerance), traced root cause to mispositioned mounting brackets, and re-verified corrected alignment before the aircraft advanced to the next station.

Avionics Systems Intern | Air India Engineering Services Limited (AIESL), Mumbai, India *Sept – Oct 2022*

- Assisted with inspection, functional testing, and recertification of 62 aircraft components across 5 MRO sub-divisions, referencing Boeing and Airbus Component Maintenance Manuals to verify compliance with design requirements.

PROJECTS

External Flow CFD Analysis, Vortex-Induced Structural Loading | *ANSYS Fluent 2024 R2, MATLAB | ASU MAE 560*

- Executed transient CFD simulations of external flow over circular and elliptical cylinders at Re = 608 across a 60-second window with 1,200 timesteps, generating velocity contours, stream function plots, and vorticity fields to characterize von Karman vortex shedding behavior and aerodynamic loading.
- Quantified that geometry orientation alone reduces oscillatory lift amplitude by up to 70% across 4 cylinder configurations, providing design guidance for minimizing flow-induced structural loading on aerodynamic components.

Independent CAD Project, NX Wing Rib | *Siemens NX, GD&T | 2024*

- Designed a mid-span wing rib for the SyberJet SJ36 proactively outside coursework, placing 8 lightening holes per Bruhn shear web criteria to achieve a 31% mass reduction (6.1 kg to 4.2 kg) under 3.8g FAR Part 23 loads — delivered as .STEP file, GD&T drawing, and written design brief.

SKILLS

CAD and Design: CATIA V5, Siemens NX, SolidWorks, Complex Surface Geometry, GD&T (ASME Y14.5-2009), Engineering Drawings, DFM, FDM Rapid Prototyping

Aerodynamics and Analysis: ANSYS Fluent (VOF Multiphase CFD, External Flow, Transient), ANSYS Mechanical (FEA, Static Structural), Hand Calculations, First Principles Sizing, Margin of Safety

Programming: Python, MATLAB (CFD Post-Processing, Data Analysis)

EDUCATION

M.S., Aerospace Engineering | Arizona State University, Tempe, AZ | GPA: 3.53/4.0 | Dec 2025

B.E., Aeronautical Engineering | Visvesvaraya Technological University (VTU), India | GPA: 3.96/4.0 | Jun 2023